Detecting Spatiotemporal Differences in Cropland Abandonment and Reforestation Across the Three-North Region of China Based on Landsat Time Series

Xuan Zhao, Taixia Wu, Shudong Wang, Kai Liu, and Jingyu Yang

***Abstract*— The Three-North region of China is an important ecological protection area. While the depopulation of rural areas and policy intervention has allowed cropland abandonment and reforestation (CAR), the spatial distribution and temporal differences underlying this landscape pattern have not been analyzed. CAR results in the gradual disappearance of the soil signal representing tilled or harvested crops, followed by a gradual increase in the interannual vegetation signal. This long-term gradual greening can be exploited to map the spatiotemporal distribution of CAR. In this study, Landsat-based detection of trends in disturbance and recovery (LandTrendr) was applied to the 1990–2020 Landsat-derived normalized difference vegetation index (NDVI) time series. The data were combined with a differential analysis of the phenological characteristics of the surveyed area to detect land-surface greening trends. The relationship with actual information on CAR was then determined. The results showed widespread long-term greening in the Three-North region, covering up to 80% of the region) during the annual growth (May–June) and harvest (September–October) periods. Together, these two periods indicated 7.5% of overall greening. The highest greening intensity was in areas where cropland had undergone conversion to forest, which strongly suggested CAR as the succession type. The average accuracy of CAR mapping was ∼80% (a mapping accuracy of a greening intensity >0.4 was as high as 87%). This study demonstrates the utility of our method in studies of driving factors of CAR and ecological benefits of secondary forest restoration.**

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***Index Terms*— Change detection, conversion of cropland to forests, LandTrendr (LTR), normalized difference vegetation index (NDVI), phenological, time series.**

I. Introduction

HUMAN land-use management is playing an increasingly important role in global land-surface greening trends [1], [2], through the intensification of agriculture but also cropland abandonment and reforestation (CAR) [3], [4]. Satellite-based time-series analyses show that agricultural land greening has accounted for nearly one-third of the greening of the northern hemisphere over the last 40 years [2] and has proceeded at more than twice the rate of greening by natural vegetation. In China, CAR is an important cause of the agricultural land greening trend, but it has yet to be evaluated in detail [5], [6].

Cropland abandonment is a global phenomenon that reflects environmental, socioeconomic, and political factors. Following the implementation of ecological restoration policies in China, a large amount of cropland has been retired and reforested in a short period of time [7], [8]. However, the environmental consequences of CAR are both positive and negative [9]. Thus, while the ecological benefits include mitigating land degradation, increasing carbon sequestration, and promoting soil nutrient cycling [7], [10], negative impacts, such as reduced surface water resources and increased risk for forest fires [11], [12], must be considered well. Previous studies on the succession of secondary forests after cropland abandonment showed that age was the most important predictor of secondary forest diversity recovery and biomass [13], [14]. Thus, mapping the distribution and timing of CAR would allow more effective and sustainable forestry management as well as agroecological policy formulation.

The Three-North region of China is an ecologically vulnerable area [15] where overgrazing and improper cultivation have caused serious land degradation and soil erosion [16]. Several ecological projects have been implemented in this region with the aim of combating the resulting desertification and sandstorms [17], [18]. For example, the aim of the Grain for Green (GFG) project, implemented in 1999, was to accelerate the conversion of cropland to forest [19]. However, due to the lack

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of effective management and long-term observational data, the spatiotemporal distribution of CAR in the Three-North region of China remains unclear [20]. In addition, the land-use/land-cover database in China is only updated every five years and the surveys do not include the categories CAR or fallow [21]. Chinese researchers have therefore often used interview-based statistics to estimate the extent of CAR at a local scale, but this approach does not provide insights into the spatiotemporal details of the process. In rural mountainous regions, the underlying driving factors of CAR are spatially heterogeneous and occur in response to the combined effects of rural demographic, socioeconomic, and policy factors [22]. Thus, CAR is typically a gradual process, often lasting for several years, and may be prolonged by interplanting until the land is no longer suitable for planting crops [23]. As a result, the intensity and pattern of CAR driven by environmental versus socioeconomic differences may differ. However, accurate information on the timing and distribution of CAR is lacking.

Vegetation status and vegetation dynamics at different scales can be effectively assessed using indices based on spectral parameters, such as the normalized difference vegetation index (NDVI) [24], [25]. Continuous vegetation development in the context of CAR appears as a gradual increase in the NDVI and can be mapped using a remote sensing dataset [26], [27]. Previously, cropland abandonment detection mapping was done using multitemporal image differences [28]. While these methods can detect cropland abandonment, they are less suitable for assessing the small-scale conversion of cropland to forest because they cannot capture the dynamic changes in vegetation over the course of a year or longer time periods.

Moreover, while remote sensing technology is an effective means of monitoring changes in agricultural land use [29], [30], its use in obtaining a detailed spatiotemporal distribution of CAR is complicated by the fact that cropland abandonment is usually spatially dispersed [5] such that changes in land-surface type may vary greatly depending on the type of abandoned crop and the subsequent succession process [31], [32]. However, the global integration of satellite imagery together with the development of cloud-based geoprocessing platforms and time-series methods has facilitated the monitoring of CAR and therefore estimations of its extent [33]. In particular, the high spatial resolution (16 m), nearly global coverage, and long-term surface records provided by Landsat satellite imagery have allowed detailed and comprehensive assessments of CAR [34], [35]. Although the efficient processing of Landsat datasets for large areas can be challenging, it has been facilitated by the development of Google Earth Engine (GEE) platforms [36].

Several recent studies have described the dynamics of cropland abandonment in detail using Landsat time series [33], [37]. For example, trajectory-based methods, such as LandTrendr (LTR) [38], can be employed to time-segment the acquired cropland probability time-series images, thus allowing an assessment of cropland abandonment [31], [32]. In this approach, the use of aggregated annual or seasonal composites is important to overcome data gap problems and effectively avoid intra-annual interference by cloud cover and rain [39], [40]. Combining the gradual increase in NDVI over time with the LTR thus offers a promising approach, as it provides a more detailed characterization of greening trends and highlights turning points in CAR events.

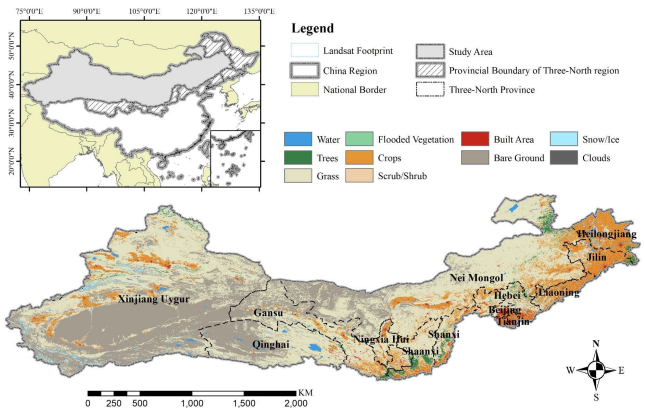


Fig. 1. Land-cover types in the study area from Sentinel-2, 10-m. Land USE/LAND COVer 2020.

Determinations of land greening in arid and semi-arid regions is an important aspect of cropland abandonment mapping [3]. Thus, in this study, we examined whether long-term gradual greening contributes to CAR mapping in the Three-North region using the NDVI time series of crops before sowing and after harvesting to estimate land greening, considering that in CAR, the soil signals generated before and after tillage are gradually replaced by vegetation signals, by analyzing greening trends and the relationship between greening intensity and CAR events, and by mapping the patterns and timing of CAR in the Three-North region.

The remainder of this article is organized as follows. In Section II, the study area is described, after which the interannual dynamics of the phenological indicators during the CAR are systematically analyzed. Then, CAR maps at the pixel level obtained from NDVI time series across multiple phenological periods using a temporal segmentation approach are presented and their accuracies are evaluated. A discussion of our findings and their implications for future research and land management concludes this report.

II. Study Area and Materials

A. Study Area

The Three-North region ranges from 34°2′ N to 50°11′ N and 73°27′ N to 128°13′ N and thus covers 42.4% of the land area in China (Fig. 1). Desertification, sandstorms, and drought threaten the region. In the early 21st century, CAR programs aimed at abandoning ecologically degraded and unsuitable croplands for reforestation were implemented nationwide, which has accelerated the conversion of cropland to forest and expanded the forested area [41], [42].

The study area has an arid and semi-arid climate, with an average annual precipitation of <400 mm. The southern part is dominated by hills and ravines, and many of its slopes have been transformed into cultivated land. The agricultural

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cycle in the study area usually starts in mid-May to mid-June, with peak vegetation growth in July and harvest from early September to late October, depending on the weather conditions. Livestock grazing is the main land use in the drier northern part of the Three-North region. Currently, croplands are mainly distributed in the northeast, which is characterized by more fertile chernozem soils. Wheat, maize, and rice are the main food crops, with only small areas planted with rapeseed, sunflower, potato, and other crops. CAR regions usually involve sloping cropland, whereas lands with higher agricultural economic benefits (such as perennial crop areas) are usually not abandoned [43].

Data on the planting area ratios of crops grown in the Three-North region were obtained from the Chinese National Bureau of Statistics (https://www.stats.gov.cn/). As shown in Table I, maize and wheat account for relatively high proportions of the area sown with grain crops. However, only 51.8% of these crops are sown in the Heilongjiang area, as rice is planted in large areas on its fertile black land. Due to the high social and economic benefits of the rice crop for the population, the province's cultivated land is rarely abandoned.

B. Data and Preprocessing

We used all available Landsat Collection 1 surface reflectance data for processing and analysis, by accessing the Landst5, Landsat7, and Landsat8 imagery archives using the GEE service. A harmonization statistical function was used to normalize the reflectance among the three sensors [44]. These data products are atmospherically corrected for clouds, cloud shadows, water, and snow using the respective masks, which can be generated or removed using the CFMask algorithm [45]. In this study, a total of 259 scenarios of the Landsat footprint coincident with the Three-North region and study period from 1990 to 2020 were used. To avoid dates with high snow probability and poor image quality, only images acquired between March 1 and November 31 and with a cloud cover of $<70\%$ were included.

Land-use product data were derived from the ESA Sentinel-2 imagery at 10-m resolution and used to distinguish vegetation types and their optical characteristics [46]. NDVI values from the MOD13Q1 product in Collection 6, obtained by MODIS onboard the Terra satellite, were used to derive vegetation properties.

III. Methods

We examined whether the conversion of cropland to forest was reflected by a gradual increase in NDVI [3] and could thus be detected by long-term monitoring of the seasonal changes in the NDVI [38]. Only the conversion of cropland to forestland was included; the transformation of cropland to grassland was excluded because it is rare in the study area, and due to the lack of rainwater resources, there is no obvious fluctuation in the NDVI time-series curve of grasslands. The study protocol consisted of five steps (Fig. 2). First, a Landsat NDVI time-series dataset from 1990 to 2020 was generated. Second, CAR areas were selected using high spatial resolution satellite imagery (Google Earth imagery; 0.6 m) and field observations

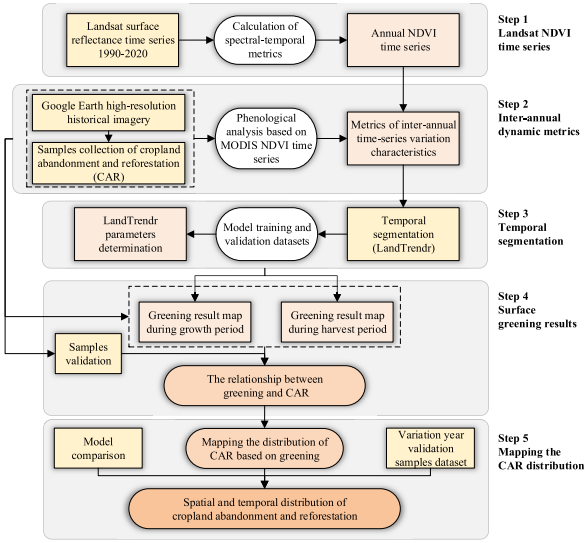


Fig. 2. Workflow used to map the distribution of CAR in the Three-North region.

data. The interannual dynamics of the phenological indicators were analyzed from the images using the MODIS NDVI time series. Third, based on the Landsat NDVI time series from 1990 to 2020, periods of gradual increase in the surface NDVI were separated using the time segmentation algorithm LTR. Fourth, the relationship between long-term vegetation greening and the conversion of cropland to forest in the study area was analyzed according to the results of the greening trend assessment and the greening intensity using the LTR. Finally, the accuracy of the results was determined and the spatiotemporal distribution of CAR in the Three-North region was mapped.

A. Generating Monthly Landsat NDVI Time Series

Nearly, all high-quality images available across the study area were used, but these images were unevenly distributed over the years, with particularly low data availability in the early to mid-1990s. While wheat and maize can be separated from grasslands in intensive management systems [31], productivity in the Three-North region is often below the maximum (e.g., due to the poor quality of the surface soils and insufficient fertilizer inputs) such that cropland may be confused with grassland. The use of images from multiple seasons is key to differentiating cropland from grassland, given that recently plowed (spring) or harvested (fall) fields can be reliably distinguished from other types of land cover [47], [48]. To compensate for data scarcity, Landsat imagery data of adjacent years for the same period were used, justified by the fact that in the arid and semi-arid regions of China, the abandonment and reforestation of farmland is usually a gradual process, and forest grasses and crops have similar growth cycles. Consequently, the spectra of ground objects in adjacent years will not significantly differ. The same strategy of using high-quality images from the adjacent years was adopted to replace low-quality images disturbed by clouds and rain [49], [50]. In this study, two seasonal periods were specified. The average NDVI at the same location was calculated

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for each period and used as a parameter in the method simulations. This reduced the uncertainty in the process of NDVI time-series reconstruction. Data compensation using data from adjacent years yielded a full Landsat time series at annual intervals, despite relatively few observations over time [33].

The NDVI was calculated and a consistent time series of NDVI spectral temporal indicators was derived based on the constructed full Landsat series. Then, the image collections for each year of the study period were subdivided into nine monthly image collections (March–November) and the mean of the spectral indices for each month was calculated. This was done based on phenology considerations: summer (July–August) was excluded because it is the time when the herbaceous vegetation is dry, but the trees are leafy. However, this period is also phenologically dynamic, which makes trend analysis more sensitive to false disturbance detections [51]. Compared to spring (March–May) or autumn (September–November), the vegetation is either just sprouting or it is withering, but the vegetation phenology is relatively stable [52]. For manually managed farmland, plowing or harvesting during spring or autumn causes significant changes in surface spectral information. The use of a finer time interval increased the sensitivity of the change detection algorithm to these and other ground disturbances.

B. Trajectory Analysis to Generate Detection Metrics

Land-cover change, depending on its intensity [53], [54], will be accompanied by sudden or gradual changes in the land-surface spectral NDVI time series, thus clearly showing the transition between cropland and noncropland [5], [32]. Remote sensing imagery at the start and end of crop growth allows cropland to be distinguished from noncrop land [47] because plowing or harvesting of the land in manually managed cropland produces clear soil signals throughout the year [8], [33]. However, with a change in land-use attributes, those signals will gradually disappear. The lack of water resources in arid and semi-arid areas causes completely different seasonal greenness signals between cropland and natural vegetation [55], [56], particularly during the initial stage of vegetation development within a CAR region, when the signals usually remain low throughout the year. In the subsequent years, the greenness signals representing the vegetation region tend to be higher than those of managed croplands during both plowing and harvesting [57], [58].

Therefore, the NDVI time series for crop tillage and harvest were determined and used as input parameters in the trajectory-based methods to detect the conversion of cropland to forest over time [38]. Fig. 3 shows the NDVI time-series curves of surface vegetation types in the study area, mainly for maize and wheat. Different growth cycles of the same crop were due to spatial heterogeneity and to differences in planting patterns [49]. The time span of the two phases was extended to two months to further reduce the impact of noise during the year. Then, the mean NDVI for each period was calculated to construct the interannual time series. Differences in crop phenology and the transition between crop types can result in erroneous change detection by the

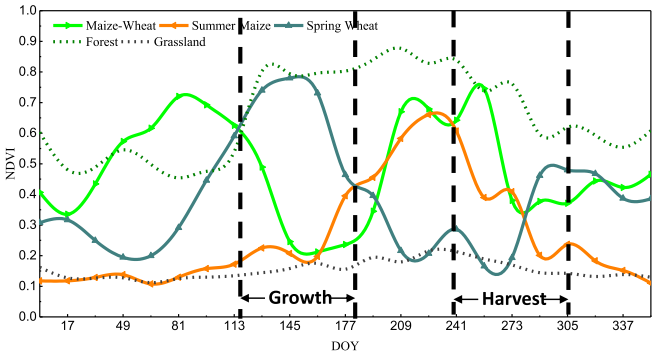


Fig. 3. NDVI time-series curves of land-surface vegetation types in the Three-North region.

TABLE I

PROPORTION OF THE AREA SOWN WITH MAIZE AND WHEAT CROPS IN THE THREE-NORTH REGION, BY PROVINCE

Province	Planting area (%): 2010-2019 average
Xinjiang Uygur	95.1%
Qinghai	61.2%
Gansu	69.6%
Ningxia Hui	80.5%
Nei Mongol	72.7%
Shaanxi	93.1%
Shanxi	89.3%
Hebei	86.2%
Liaoning	76.7%
Jilin	75.5%
Heilongjiang	51.8%

trajectory-based method. In this study, the changes detected using the trajectory-based method for the NDVI time series should have been the same for the May–June growth period (GP) and the September–October harvest period (HP), due to the similar greening trends. The latter reflects the fact that when arable land planted with wheat is replanted with maize, the average NDVI parameters in GP decrease significantly and those in HP increase significantly. When arable land planted with maize is replanted with wheat, the average NDVI parameters in GP significantly increase and those in HP significantly decrease.

LTR implemented in GEE was applied to the temporal trajectory of GP and HP [38], [59]. The use of GEE avoided the problem of the enormous amount of time and storage space needed to download and process the data [60], [61]. The LTR algorithm fits spectral trajectories on a pixel-by-pixel basis, using regression methods for point-to-point fitting to obtain annual time series [51]. Among the metrics that can be derived based on these time series are the number and length of line segments (in years), their slopes, and the years corresponding to the vertices of the line segments. In this study, LTR was run through a set of parameters by visually inspecting the CAR samples identified by Google Earth high-resolution imagery. Changes in the trajectory fit over these sample regions in response to the different parameters were visually assessed and the parameter fit across samples that resulted in the best time-series segmentation was identified [31]. As shown in Table II, maximum segmentation,

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TABLE II
PARAMETER VALUES FOR LTR USING THE GEE

Parameter	Value
maxSegments	6
spikeThreshold	0.9
vertexCountOvershoot	3
preventOneYearRecovery	True
recoveryThreshold	0.25
pvalThreshold	0.05
bestModelProportion	0.75
minObservationsNeeded	6

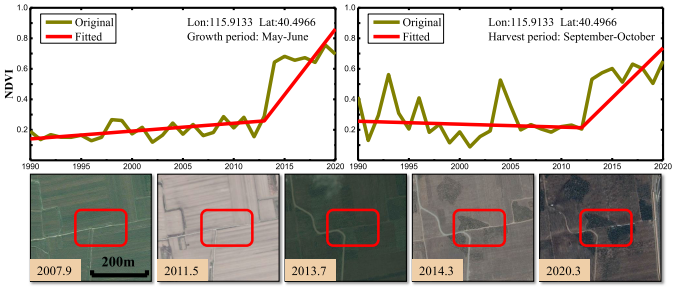


Fig. 4. Example of land-use change due to CAR, as shown in (bottom) Google Earth high-resolution satellite imagery and in time series of the mean NDVI values calculated over (top left) May–June and (top right) September–October. Brown line: average NDVI value for the respective year. Red line: LTR segments. Red border: data collection location.

despiking, and the recovery threshold allowed fallow land to be separated from the long-term retirement signal, as otherwise one or two fallow years may be wrongly identified as land retirement. This parameterization was applied to map both gradual and sudden greening increases during GP and HP from 1990 to 2020 (Fig. 4). The changes were considered statistically significant at $p < 0.05$. The final results showed the magnitude of the NDVI increase and the year in which the change occurred. Saplings are usually planted between spring and summer, to increase their survival rate. Therefore, this study used the time results of LTR detection at HP to generate the final temporal distribution map of CAR.

C. Testing the Relationship Between Greening and the Conversion of Cropland to Forest

Information on greening and the individual greening intensity determined from the CAR samples based on the LTR detection algorithm was exported and used to study the relationship between greening and CAR events. In addition, the average NDVI values of the CAR samples at the start (1990), middle (2005), and end (2020) of the study period were calculated to compare and analyze the NDVI distribution of managed farmland and CAR regions during different periods.

The magnitude of change in the NDVI greening trend was assumed to be an indicator of a CAR event. Locations in which the magnitude of change was above or below the threshold were classified as retired cropland or nonretired cropland, respectively. To explore the relationship between greening intensity and cropland conversion to forest, the minimum greening threshold was iteratively increased from 0 to 0.4 in steps of 0.1. Greening thresholds >0.4 were grouped into one category. Disproportionately stratified sampling was used to

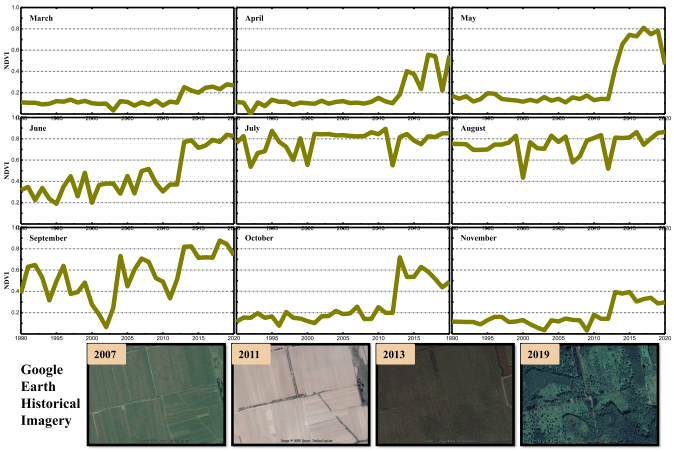


Fig. 5. (Bottom) Google Earth high-resolution time-series images of CAR events and (top) corresponding monthly Landsat NDVI time series.

collect 200 samples from each of the five greening intensity intervals [62], resulting in a total of 1000 samples. Then, the five groups of samples were compared to land-use information intuitively interpreted from the Google Earth imagery. CAR event detection was categorized as correct, nondetected, or misclassified. The producer accuracy (PA), user accuracy (UA), and overall accuracy (OA) were calculated from a confusion matrix to test the accuracy of the CAR events. Validated samples of abandoned cropland were obtained from high-precision historical Google Earth images. The reduction in cropland area between years was directly obtained by visual interpretation, and the transition from farmland to woodland was defined as a decrease in crop coverage and an increase in forest area both exceeding 50% of the pixel range.

Our approach was used to map the distribution of CAR events during the study period. A CAR event was considered to have permanently changed the attributes of land use. In contrast to a natural succession of vegetation that does not lead to land-use change, the greening tendency is generally weak. Therefore, in the analysis and verification of the time point of CAR event occurrence, verification was limited to areas with a high (>0.2) greening intensity. TimeSync was used to explain the time trajectory at the sample position and, thus, the temporal dynamics in the time series [59]. A second sample dataset, including 300 samples, was also collected, in which 231 samples were successfully explained and 69 samples were discarded due to uncertainty. Considering the gradual spectral variation in the cropland abandonment area and thus the complexity of visual identification, a fuzzy validation method allowing a confidence interval of ±1 year was applied. Based on the time results detected in this study, PA, UA, and OA were calculated through a confusion matrix to test the accuracy of occurrence time. This was done only for CAR events between 1992 and 2018, as the two years before and after a time-series change detection are typically unreliable [60].

IV. RESULTS

A. NDVI Time-Series Characteristics of LULCC

Fig. 5 shows the monthly Landsat NDVI time series of the CAR region. According to the NDVI time series

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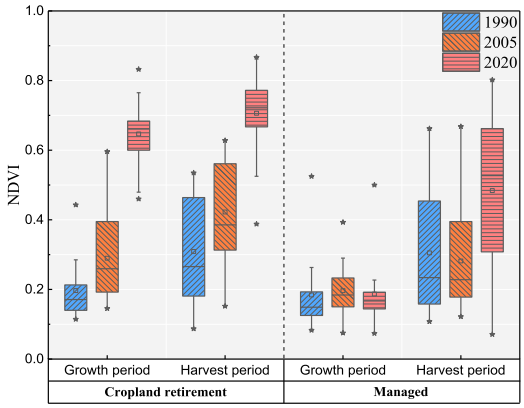


Fig. 6. Distribution of the mean NDVI values of different land-use patterns at the start (1990), middle (2005), and end (2020) of the study period.

from 1990 to 2012, stable cropland is characterized by a low NDVI value maintained during the early stage of crop cultivation, with an average of 0.1–0.2. Crops are planted from mid-May to mid-June, depending on the temperature and rainfall conditions, and harvested from early September to late October, at which time the NDVI value remains at ∼0.2. For vegetation that grows continuously, as assessed using the NDVI time series from 2012 to 2020, forest and grass growth begin in early spring (March) and are indicated by a low NDVI value (∼0.3). With the arrival of the growing season (June–August), the continuous growth of natural vegetation and stable cropland is indicated by NDVI values that are initially high but gradually decline until autumn (September–November). Changes in land-use attributes (such as the conversion of cropland to forest) change the spectral timing characteristics of the land-cover types. In this study, compared to the nonsignificant change in the annual NDVI maximum composite, prolongation of the growth cycle was the main phenological change that characterized CAR. In areas of CAR occurrence, the monthly NDVI values of May–June and September–October were significantly different, with an average difference as high as 0.7 and 0.4, respectively (the specific value depended on the vegetation coverage).

Based on reference sample information on CAR events and managed cropland, the greening trend was found to impact both managed cropland and CAR areas (Fig. 6). In areas of conversion from cropland to forest, the average NDVI value at HP in any year (1990, 2005, and 2020) was higher than that at GP and showed the same significant increasing trend. Within the scope of managed cropland, the mean NDVI values of GP fluctuated around 0.15 in 1990, 2005, and 2020 and then remained relatively stable. However, during HP, particularly in 2020, the average NDVI value significantly increased to ∼0.5. During GP (May–June), the NDVI time series of the CAR area significantly increased compared to that of managed cropland. During HP (September–October), the NDVI time series in the CAR area also significantly increased.

B. Mapping the Distribution of CAR Based on Greening Estimated by LTR

Fig. 7 shows the increases in the NDVI values in the Three-North region as well as the differences in those values

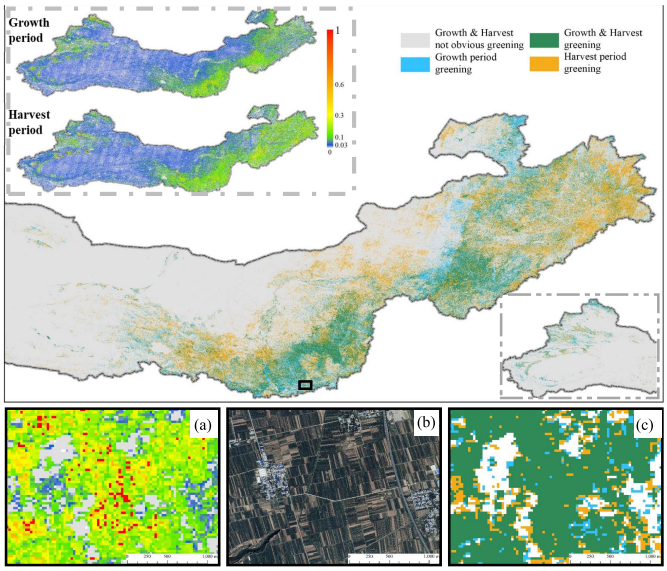


Fig. 7. Greening regions according to LTR greening masks during the GP and HP. The black rectangle is the area shown below the main map. (a) Greening results (using GP as an example). (b) Google Earth high-resolution image. (c) High-intensity greening distribution when GP and HP were considered concurrently.

depending on the phenological stage. According to the LTR method, during GP, 85.9% of the NDVI values indicated a significant and continuous greening trend and 14.1% of the NDVI values had either no significant trend or declined. High-intensity greening (NDVI value > 0.1) occurred in only 12.4% of the study area. During HP, gradual greening occurred over 83.2% of the study area, with high-intensity greening accounting for 16.7%. When the two periods were considered together, high-intensity greening occurred in 7.5% of the study area.

An analysis of the greening trend during GP showed that extensive high-intensity greening was mainly concentrated in Shaanxi, Shanxi, and Hebei provinces as well as in northeastern Inner Mongolia and northern Xinjiang. Central and western Inner Mongolia and most of Xinjiang had a low greening intensity. Analysis of the greening trend during GP showed extensive high-intensity greening also in central Inner Mongolia and areas, including Heilongjiang, Jilin, and Liaoning. When both GP and HP were considered, high-intensity greening was concentrated in Shaanxi, Shanxi, and Hebei provinces. A single high-intensity greening trend, attributable to the development of agricultural intensification and changes in crop types, such as winter wheat, was detected in Heilongjiang, Jilin, and Liaoning at HP, as well as in southern Gansu.

There were obvious differences in the number of detected CAR events during different greening intensity intervals. During GP, <20% of the CAR events had a greening intensity of 0–0.1 whereas >78% of the CAR events had a greening intensity of >0.2. During HP, >85% of the CAR events had a greening intensity of >0.2. Based on the greening intensity during both periods, i.e., >0.2 and >0.1, the accuracy of detecting a CAR event was >90% and ∼60%, respectively (Fig. 8). Therefore, in this study, a greening intensity threshold of 0.1 was used to distinguish CAR from non-CAR events.

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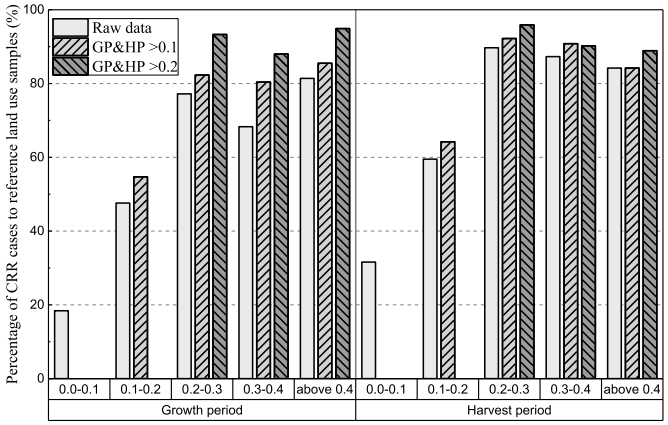


Fig. 8. Number of CAR events detected in the greening intensity interval (%).

TABLE III
ACCURACY ASSESSMENT OF THE SPATIAL AND TEMPORAL DISTRIBUTIONS OF CAR BASED ON LTR ESTIMATION OF GREENING INTENSITY

Greening intensity interval	Class (pixels)	Cropland abandonment and reforestation	Other Land use (OLU)	User's accuracy
0.1-1	CAR	438	98	81.7%
	OLU	63	201	76.1%
	Producer's accuracy	87.4%	67.2%	Overall accuracy 79.8%
0.1-0.2	CAR	39	36	52.0%
	OLU	26	99	79.2%
	Producer's accuracy	60.0%	73.3%	Overall accuracy 69.0%
0.2-0.3	CAR	106	28	79.1%
	OLU	21	45	68.9%
	Producer's accuracy	83.5%	61.6%	Overall accuracy 75.5%
0.3-0.4	CAR	120	24	83.3%
	OLU	13	43	76.8%
	Producer's accuracy	90.2%	64.2%	Overall accuracy 81.5%
0.4-1	CAR	136	17	88.9%
	OLU	9	38	80.1%
	Producer's accuracy	93.8%	69.1%	Overall accuracy 87.0%
Time Detection	**Class (pixels)**	**LandTrendr (LTR)**	**TimeSync (TS)**	**User's accuracy**
	LTR	144	15	90.6%
	TS	23	49	68.1%
	Producer's accuracy	86.2%	76.6%	Overall accuracy 83.5%

Confusion matrices were calculated for the overall detected number of CAR events. Verification was carried out when the greening results of the two periods had the same greening intensity interval. A confusion matrix was also calculated for the number of CAR events in different greening intensity intervals. Table III shows the high classification accuracy within an interval of 0.1–1, with PA, UA, and OA values of 87.4%, 81.7%, and 79.8%, respectively. The OA in the detection of a CAR event with greening intensity intervals of 0.1–0.2, 0.2–0.3, 0.3–0.4, and 0.4–1 were 69.0%, 75.5%, 81.5%, and 87.0%, respectively. There was a clear positive correlation between the greening intensity and a CAR event. A high accuracy was also determined for the occurrence time of a CAR event as detected using the method described here, evidenced by an OA of 83.5%.

Confusion matrices of the global land-cover (GLC) products were constructed based on historical Google Earth images (Table IV). In addition, land-use type was determined by support vector machine using multidate images. The mapping accuracy of the conversion of cropland to woodland obtained

TABLE IV
CONFUSION MATRICES OF VARIOUS MODELS BASED ON GOOGLE EARTH IMAGES

Model	Class (pixels)	CAR	Other changes	User's accuracy
$LTR_{greening}$	CAR	498	106	82.45%
	Other changes	93	334	78.22%
	Producer's accuracy	84.26%	75.91%	Overall accuracy 80.70%
$GLC_{transfer}$	CAR	326	117	73.59%
	Other changes	198	390	66.33%
	Producer's accuracy	62.21%	76.92%	Overall accuracy 69.45%
$SVM_{transfer}$	CAR	312	127	71.07%
	Other changes	146	446	75.34%
	Producer's accuracy	68.12%	77.84%	Overall accuracy 73.52%

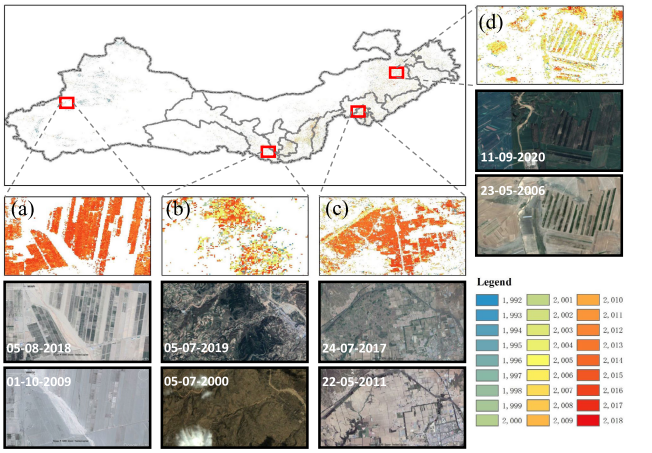


Fig. 9. CAR event detection map based on the LTR algorithm and the Google Earth high-resolution satellite imagery of the corresponding year. (a)–(d) are partial enlarged images of CAR event detection results, respectively.

with this approach was lower than that of our method, although the accuracy of CAR detection was good. Nonetheless, the highest classification accuracy was achieved using the method developed in this study, as demonstrated by UA, PA, and OA values of 82.45%, 84.26%, and 80.70%, respectively. Although the OA of the GLC and SVM products was 69.45% and 73.52%, respectively, the PA was only 62.21% and 68.12%, indicating that nearly half of the cropland abandonment results were misidentified.

C. Temporal and Spatial Patterns of CAR

Fig. 9 shows the different spatial patterns in the conversion of cropland to forest in the Three-North region. CAR events between 1992 and 2018 usually developed gradually. For example, large-scale CAR events in southern Gansu, northern Shaanxi, western Shanxi, and eastern Inner Mongolia were characterized by scattered and irregular spatial modes, with the conversion of cropland to forest land completed slowly [see Fig. 9(b) and (d)]. In contrast, continuous patches of CAR mainly occurred in Hebei Province, in central and western Xinjiang, and particularly in northwestern Beijing. In these regions, CAR events developed mostly within a short time period [see Fig. 9(a) and (c)]. The spatial patterns of CAR ranged from larger, continuous patches to more scattered, irregular patterns, with the differences and diversity together suggesting different disturbance factors.

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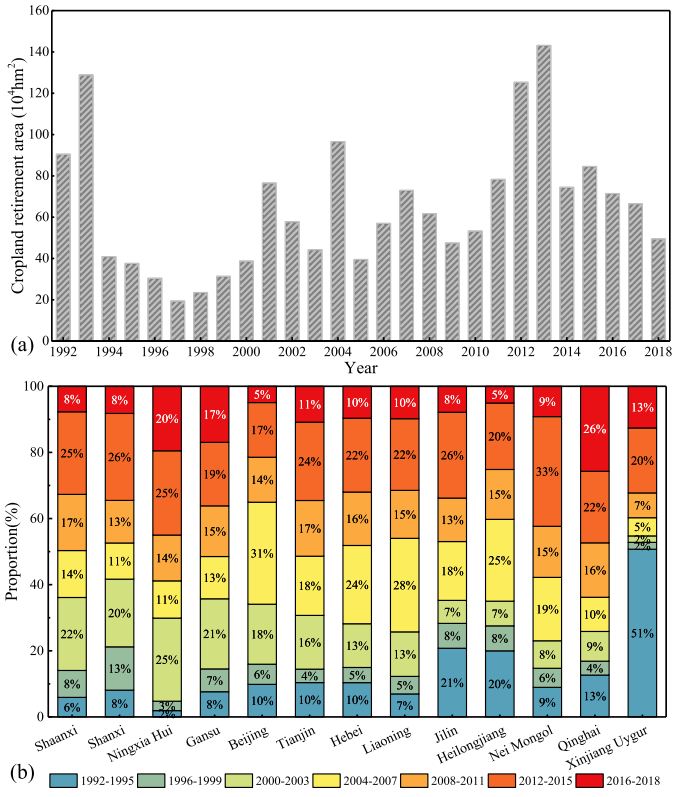


Fig. 10. Analysis of the dynamics of CAR: 1992–2018. (a) Annual CAR area estimated from Landsat composites. (b) CAR area as a percentage of the total area.

Careful observation and analysis of the timing distribution, based on the simulation results of the LTR method using a combination of the spectral index and monthly data (NDVI; May–June and September–October), provided further insights into the spatiotemporal patterns of CAR across the study area. CAR events in Xinjiang mainly occurred at the beginning of the observation period (Fig. 9) and mostly involved artificially planted shrubs. In the south-central part of the study area dominated by ravines and hills, the changes were usually smaller and irregular, mainly occurred from 2000 to 2005, and mostly involved slope cropland in mountainous areas. A larger change occurred around 2015 and was relatively concentrated, as it mainly involved the flat cropland region.

Estimation of the annual area of CAR events over time in the Three-North region showed changes from year to year. The area of CAR exceeded 1.2 million hm^2 in 1993 and 2012 and reached a peak in 2013, with an area of 1.432 million hm^2. From 1994 to 2000, the observed area of CAR was generally small, around 3×10^5 hm^2. The total area of CAR across the entire study period was 17.42 million hm^2 [Fig. 10(a)]. From 1992 to 2018, there were clear interannual differences in the percentage of the CAR area in each province [Fig. 10(b)]. In Shaanxi, Shanxi, Ningxia, and Gansu provinces, CAR was first implemented on a large scale from 2000 to 2003, at which time the average proportion of CAR exceeded 20%. The second large-scale CAR effort occurred after 2012. In Beijing, Tianjin, and Hebei, CAR mainly occurred between 2004 and 2015, and in Liaoning, Jilin, Heilongjiang, and Inner Mongolia, CAR mainly occurred around 2015. In Xinjiang, it generally occurred in the early 1990s, and in Qinghai, it occurred around 2015.

V. Discussion

Previous studies that have estimated the contribution of the GFG project to greening rarely have used land greening as an indicator of CAR. In this study, this issue was explored by examining the relationship between land greening trends and the conversion of cropland to forest. We found a gradual increase in NDVI during GP and HP, reflected in the gradual development of continuous vegetation on unmanaged cropland and detectable by the long-term monitoring of NDVI changes. In addition, combining the greening trends during GP and HP with Landsat monthly NDVI time series can be used to map the spatial and temporal distribution of CAR.

A. Linking Changes in Land Use and Cover to NDVI Trends

Time-series methods have attracted increasing research interest in the study of landscape classification and landscape changes [38], [63], [64], [65]. The trajectories of time-series data from a specific type of change, such as fire, logging, or reforestation, or from multiple changes have been analyzed with the aim of detecting gradual or sudden changes in land cover [63], [64]. In this study, the time segmentation method LTR was used to segment the greening trend of different phenological stages of the surface vegetation as determined using NDVI time series. Based on the differences in the phenological characteristics of crops and forest, highly accurate information on both the spatial distribution and time course of the conversion of cropland to forest could be obtained. The use of GEE resolved the limitations posed by the need to process massive amounts of data and enabled efficient mapping of the temporal and spatial distribution of cropland converted into forest, consistently, and on a wide regional scale.

According to the NDVI trends, >80% of the Three-North region in China underwent greening (both low and high intensities) during May–June and September–October of the study period. However, high-intensity greening during both GP and HP accounted for only 7.5% of the study area and was concentrated in the southeast. In the central and western regions of the study area, characterized by a drier climate, high-intensity vegetation greening was not observed, consistent with previous research [66]. China has implemented a number of ecological restoration projects in the Three-North region, and their positive impact has been demonstrated in several studies [67]. Yet, other studies have suggested that large-scale reforestation efforts have had limited success in reducing desertification [68], particularly in the arid and semi-arid areas of northern China, where the overall survival of the trees in afforestation projects is only 15% and is limited to trees planted in wetter areas [69].

In the range of managed cropland in our study area, the interannual NDVI greening trend during May–June (GP) was not obvious, unlike the obvious upward trend in September–October (HP). In addition to the impact of rainfall and temperature changes [70], [71], this positive trend of NDVI in the latter can be explained by the change in the

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main crops together with agricultural intensification. Because changes in planting patterns and planting structures can cause sudden or gradual changes in the interannual NDVI trends, the NDVI greening trends in the two phenological stages were simultaneously analyzed to reduce the interference of similar land-cover changes on the results of the model.

B. Cropland Abandonment and Reforestation

An important finding of this study was the more intensive greening due to CAR than to manage cropland in the Three-North region. In this study, vegetation dynamics during the GP and HP of the arid and semi-arid study area were monitored for changes in land cover based on interannual NDVI time-series analysis. The results showed that the transition from cropland to forest was accompanied by the highest greening trend and the highest greening intensity, strong evidence of CAR. Usually, ecological restoration policies include the planting of vegetation shortly after cropland abandonment such that the development and densification of shrubs and trees lead to a high rate of greening [3]. However, the relationship between greening and the conversion of cropland to forest must be considered from other perspectives as well because cropland abandonment and the subsequent increase in vegetation growth are only one of many drivers of greening.

Although the NDVI-determined greening intensity during GP and HP indicated the occurrence of CAR, obvious greening was also seen in some managed agricultural land (Fig. 8), presumably due to agricultural intensification [2], [72]. In addition, cropland retirement and reforestation may take several years and may be interrupted by the interplanting of crops and shrubs [8] such that a complete change in the spectral characteristics of the land surface may take place slowly. These issues pose challenges for setting accurate thresholds of greening intensity to clearly distinguish land conditions.

In our approach, both impervious surfaces and water bodies were masked to ensure that CAR detection was confined to cropland and grass pixels. Nonetheless, omissions and erroneous results were also present and can partly be explained by the relatively broad range of crop types and tree categories. When the results of high-intensity afforestation during GP and HP were used to identify CAR events, then from the 1990s to the end of the study period, $\sim$17.42 million hm^2 of land in the study area was determined to be covered with continuous vegetation. Furthermore, given that most of the recent conversion of cropland to forest does not yet show a clear continuum, the actual area may be larger.

The CAR events detected in this study coincided with the dates of policy changes in China [4], [73]. Thus, large-scale greening was first detected around 2000, 2007, and 2014, which corresponded to the period when ecological restoration policies were implemented in China. In 1999, the Chinese government carried out a GFG pilot project in Gansu and Shaanxi and then extended it to 21 other provinces across China by the end of 2001, which led to the afforestation of a large amount of sloping cropland in Shaanxi, Shanxi, and Gansu provinces in 2000–2003 [7]. Challenges related to the payment of state subsidies to cropland owners complicated the implementation of the GFG project after 2004, with a decrease in the conversion rate occurring after 2005. To ensure food security, in 2007, the Chinese government proposed a red "safety" line of no less than 1.8 billion mu of arable land and raised the subsidy standard for planting between 2007 and 2013. In 2014, the government proposed a second round of GFG [74]. Our results show that the occurrence of the CAR events followed the same trend as the implementation of ecological restoration policies.

C. Applicability and Limitations of the Method

This study demonstrated that our proposed method of CAR assessment can be applied in relatively arid regions, where the impact of missing key data due to cloud and rain interference is small and the cropping system is usually carried out only once a year [75]. The monthly synthetic data captured changes in the spectral signatures caused by land-cover changes. Our method proved to be very effective for monitoring the spatiotemporal distribution of the GFG project across the Three-North region. Moreover, because it uses both the GEE platform to access the Landsat collection and high-performance computing services, the developed method can be easily modified to allow its use in other study locations and during other annual periods. Furthermore, the LTR method enables the successful detection of gradual as well as sudden changes in a time series.

Time-series segmentation allows the capture of patterns representing the conversion of cropland to forest. Although in this study, interannual time series of land-cover change were evaluated considering all data from the GP and HP, the specific phenological characteristics of different crops were not considered. Rather, maize and wheat crops in the study area were considered to be representative and were used to systematically analyze the interannual differences in the phenological characteristics of two major agricultural periods and thus to detect the spatial and temporal distribution of CAR in the Three-North region. However, determinations of the spatiotemporal distribution of CAR on a large scale with high recognition accuracy will require systematic analyses of the phenological differences between crops of different types and between crops and other vegetation types.

The ultimate goal of returning cropland to forest is forest regeneration, although even with artificial planting, forest regeneration is a gradual process. However, regeneration is rarely assessed in a continuous fashion using remote sensing, despite its value in systems with strong human-induced changes in land cover, where the continuous detection of forest changes is essential. Our approach can provide information on the distribution and timing of GFG, while the results related to greening intensity will improve assessments of the restoration potential of secondary forests [76].

VI. Conclusion

The method proposed in this study uses greening intensity as an index to detect the conversion of cropland to forest, thus providing information on the distribution and timing of CAR on a large scale. Analyses of time series of dense images showed a high degree of overlap between a higher

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greening intensity and the conversion of cropland to forest, a finding that largely depended on the use of appropriate phenological analysis methods. Our results also showed that long-term greening was very common in the study area, with the greening area accounting for >80% during GP and HP. This large-scale greening across the study area reflected the implementation of an ecological restoration policy in 1999. Dense, high-intensity greening was shown to occur on agricultural land that had ceased to be cultivated, which implies that cropland owners had intentionally transformed cultivated land into green land, driven by government efforts at ecological restoration. According to our estimate, during the study period, >17.42 million hm^2 of land in the Three-North region underwent continuous revegetation. However, the restoration potential of secondary forests after cropland abandonment could not be similarly assessed due to the complex changes in the spectral trajectories of the surface vegetation, which indicated gradual, abrupt, but also stable changes. Obtaining large-scale information on the distribution and timing of the conversion of cropland to forest will be a key aspect of further research on the factors driving this conversion and the benefits of ecological restoration policies.

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